8_apr_

08 April 2026 20:18

☆ Regression

Supervised learning.

↳ data set → linear regression.

one

mul

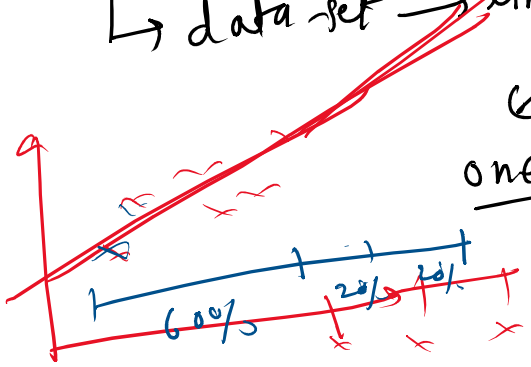
logistic regression.

① ⇒ conditional prob?

② ⇒ likelihood?

③ ⇒ entropy?

④ ⇒ cross entropy?



⇒ input $x \longrightarrow y$

$$y = \beta_0 + \beta_1 x + \varepsilon$$

↳ R^2

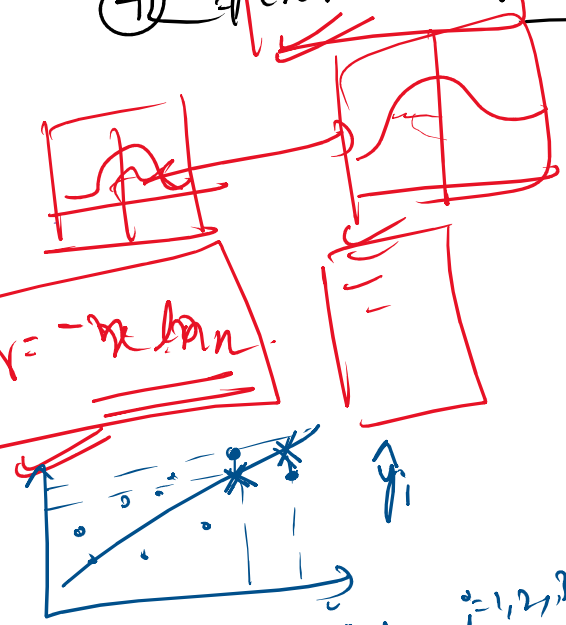
↳ MAE

↳ SE

↳ MSE

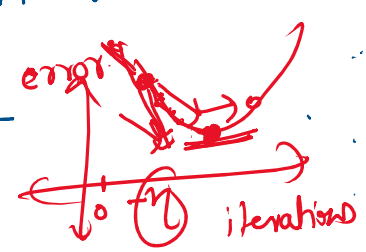
$$\frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_i - \bar{y}\right|^2$$

$\boxed{\text{entropy} = -x \log n}$

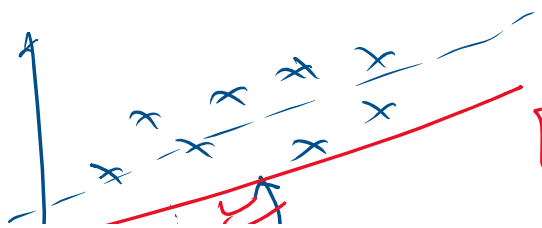


y_i $\quad \sum_{i=1}^{n} |y_i - \hat{y}_i|$ $\quad i = 1, 2, 3, \ldots$

objective ⇒ $\min\left[J(\beta_0, \beta_1)\right] = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$



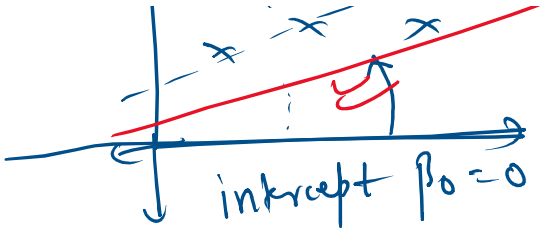
$y = 0$



We need to min. (J).

$[\beta_0 \quad \beta_1]$.

↳ (optimized)



intercept $\beta_0 = 0$

slope $\beta_1 = 0$

$[\beta_0 \quad \beta_1]$

gradients: (optimizer)

$$\frac{\partial J}{\partial \beta_0} = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)$$

$$\frac{\partial J}{\partial \beta_1} = \frac{-2}{n} \sum (y_i - \hat{y}_i)$$

update rule:

$$\beta_0 := \beta_0 - \eta \frac{\partial J}{\partial \beta_0}$$

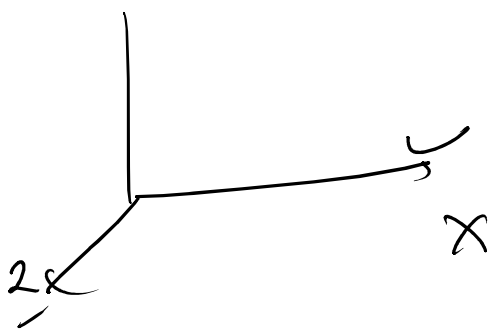
$\eta \to 0.1 \,/\, 0.01$

$$\beta_1 = \beta_1 - \eta \frac{\partial J}{\partial \beta_1}$$

$\eta \to$ learning rate.

* multiple inputs

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_n x_n$$



$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ \vdots & & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

$m \to$ no. of data points.

$n =$ dim.

$n = \dim$

$X \in \mathbb{R}^{n \times (p+1)}$, $\beta \in \mathbb{R}^{(p+1)\times 1}$

$y \in \mathbb{R}^{n \times}$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix} \quad \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

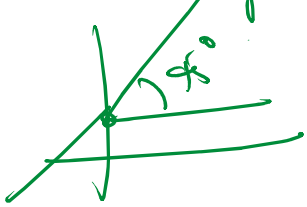
$$J(\beta) = \frac{1}{n}(y - X\beta)^T (y - X\beta)$$

$(y - \hat{y}_i)^2$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$y = x + 1$



$$\beta = (X^T X)^{-1} X^T y$$

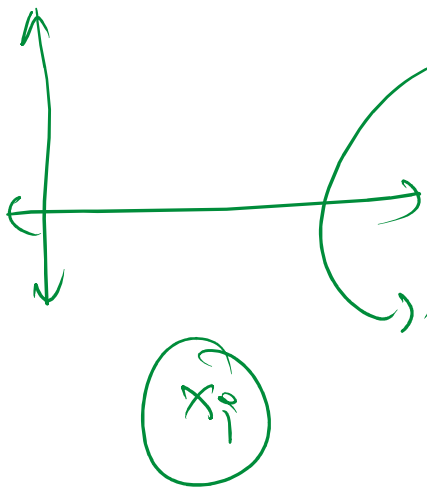
gradient descent

$$\nabla J = \frac{-2}{n} X^T (y - X\beta)$$

$$\beta := \beta - \alpha \nabla J$$

$\alpha \rightarrow$ learning rate

* polynomial regression?



$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d$$

$$z_1 = x, \quad z_2 = x^2, \quad \ldots \quad z_d = x^d$$

$$y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \cdots + \beta_d z_d$$

↳ linear in z.

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_1 & x_2^2 & & x_2^d \\ 1 & x_1 & x_3^2 & & \vdots \\ \vdots & & & & \\ 1 & x_1 & x_n^2 & & x_n^d \end{bmatrix} \quad \beta = \begin{bmatrix} 1 \\ \beta_1 \\ \vdots \\ \beta_d \end{bmatrix}$$

$$\hat{y} = X\beta$$

$$J(\beta) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\beta := \beta - \underbrace{\alpha}_{\text{l.r}} \left(-\frac{2}{n} X^T (y - X\beta) \right)$$